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MATLAB 6

Exercise

Consider the initial value problem:

$$\begin{aligned}\frac{dx}{dt} &= -2x + \frac{1}{2}y \\ \frac{dy}{dt} &= -y\end{aligned}, \quad x(0) = m, \quad y(0) = n.$$

% Can be rewritten as $x' = -2x + 1/2y$ and $y' = -y$

1. Draw a direction field for the system.

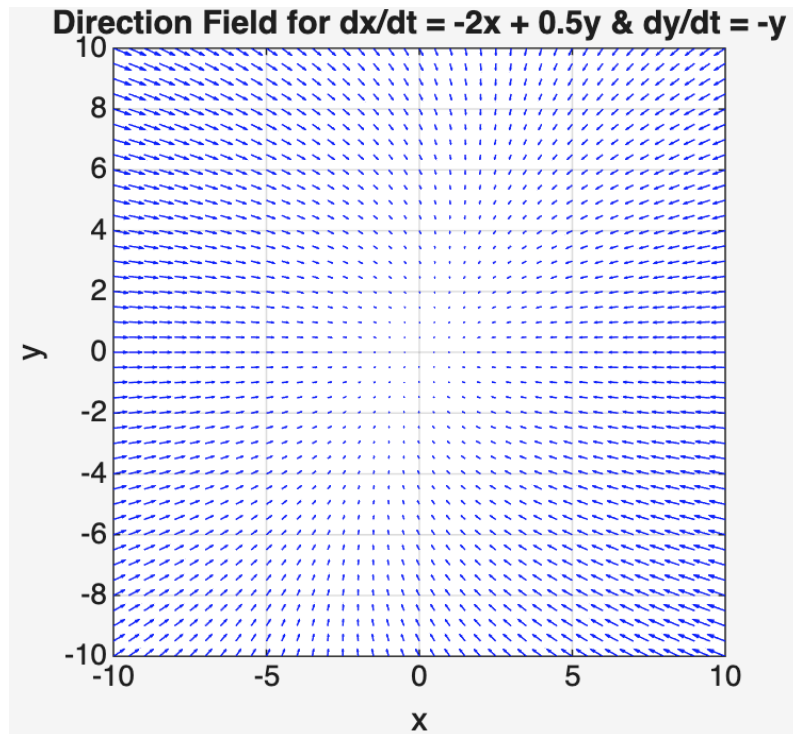
Command Window Screenshot:

Command Window

```
>> [x, y] = meshgrid(-10:0.5:10,-10:0.5:10);
>> dx = -2 * x + 0.5 * y;
>> dy = -y;
>> figure
>> quiver(x, y, dx, dy, 'b')
>> xlabel('x')

>> ylabel('y')
>> title('Direction Field for dx/dt = -2x + 0.5y & dy/dt = -y')
>> axis equal
>> axis tight
>> grid on
>> hold on
```

Direction Field Screenshot:



2. Determine the type of the equilibrium point at the origin.

Command Window Screenshot:

```
Command Window
>> A = [-2 1/2; 0 -1]

A =

    -2.0000    0.5000
         0    -1.0000

>> [V, D] = eig(A);
>> eigvalues = diag(D)

eigvalues =

    -2
    -1
```

% Both eigenvalues are real and the same sign (negative), so the equilibrium point is a stable node.

3. Use `dsolve` to solve the IVP in terms of `m` and `n`.

Command Window Screenshot:

```
Command Window
>> syms x(t) y(t) m n
>> equations = [
    diff(x,t) == -2 * x + 1/2 * y
    diff(y,t) == -y
];
>> initconds = [
    x(0) == m
    y(0) == n
];
>> Soln = dsolve(equations, initconds)

Soln =

    struct with fields:

        y: n*exp(-t)
        x: (n*exp(-t))/2 + exp(-2*t)*(m - n/2)

>> xSol = Soln.x

xSol =

(n*exp(-t))/2 + exp(-2*t)*(m - n/2)

>> ySol = Soln.y

ySol =

n*exp(-t)
```

4. Find all straight-line solutions.

Command Window Screenshot:

```
Command Window
>> A = [-2 1/2; 0 -1];
>> [V, D] = eig(A)

V =

    1.0000    0.4472
         0    0.8944

D =

    -2     0
     0    -1

>> solnslopes = V(2,:) ./ V(1,:)

solnslopes =

     0     2
```

5. Plot the straight-line solutions together with the solutions with initial conditions $(m, n) = (2, 1), (1, -2), (-2, 2), (-2, 0)$

Command Window Screenshot:

```
Command Window
>> f = @(t, z) [-2 * z(1) + 0.5 * z(2); -z(2)];
>> timespan = [0 5];
>> initconds = [ 2  1;
                1 -2;
                -2  2;
                -2  0];
>> figure
>> hold on
>> grid on
>> axis equal
>> axis tight
>> xlabel('x')
>> ylabel('y')
>> title('Phase Portrait with Straight-Line Solns')
>> for k = 1:size(initconds,1)
    [t, z] = ode45(f, timespan, initconds(k,:));
    plot(z(:,1), z(:,2), 'LineWidth', 1.8)
end
>> plot([-4 4], [0 0], 'r--', 'LineWidth', 2)
>> plot([-4 4], [-8 8], 'b--', 'LineWidth', 2)
>>
```

Plot Screenshot:

